Tarun Joseph AI Engineer · AI Agents & Evaluation

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TECHNICAL SKILLS

Languages & Tools: Python, SQL, REST APIs, Git, Docker, FastAPI, Streamlit/Gradio **| Agents:** tool calling, multi-agent orchestration, agentic RAG, knowledge graphs, MCP, LangChain, OPA policy rules, guardrails **| AI Governance:** OWASP LLM Top 10, MITRE ATLAS, EU AI Act, OSFI E-23 **| Fine-tuning & Evaluation:** QLoRA (Unsloth), Hugging Face, PyTorch, macro-F1, Cohen's kappa **| Cloud:** GCP, AWS, Azure, CI/CD

EXPERIENCE

❖ **Machine Learning Associate | Vector Institute × SafeAlign AI (Fastlane Applied Projects)** *Aug 2026 – Present*

- Building **runtime monitoring and alignment for production multi-agent systems** at financial clients, on-prem and in the cloud: OPA policy rules, a knowledge graph and **agentic RAG** over each client's policies give every agent step a PASS, ESCALATE or QUARANTINE verdict.
- Built the **threat taxonomy and crosswalk** behind the knowledge graph: merged 4 datasets' threat vocabularies into one taxonomy and linked each threat to controls, regulatory requirements, OWASP LLM and MITRE ATLAS.
- **Fine-tuned two SLMs with QLoRA** as the system's checks: one names the attack in a flagged tool output, the other screens user prompts for injection, both tested on attacker goals unseen in training.
- Built training data with an **LLM labeller**, with splits grouped so no attacker goal or customer appears in both train and test, and found and fixed label leakage in an earlier dataset.

❖ **AI Program Design & Delivery Intern | Vector Institute** *May 2026 – Aug 2026*

- Led the **Agentic AI Forecasting & Evaluation bootcamp** (**9.0/10** user rating), unblocking industry teams on AI agent use cases for CIBC, PwC and Roche.

❖ **Machine Learning Engineer | Tata Consultancy Services (TCS)** *January 2023 – May 2025*

- Deployed GPT-3.5- and Llama 3-powered **RAG chatbots and AI agents** on TCS's internal EdTech platform, serving **600,000 associates** globally.
- Built a **Llama 3 agent from PoC to MVP** that automated repetitive support tickets from TCS's EdTech learning platform, plus a chatbot developers used to close user tickets (tool calling, vector-store memory), saving each developer **5+ hrs/week**.
- Built a web-scraping and **OCR data-extraction tool**, and led GenAI bootcamps for 1000+ associates.

PROJECTS

❖ **MineMentor: Multi-Agent Minecraft Tutor | LLM Agents, Mineflayer, UofT ECE1786** *Oct 2026 – Present*

- Team project: a **chat agent** turns player requests into a task queue, an **orchestrator** runs them on a Mineflayer game agent, and a **rule-based detector** reads gameplay to decide when to hint or stay quiet.
- Own **calibration, data labelling, evaluation and testing**: sessions are cut into 30 s windows labelled blind from screen recordings, with Cohen's kappa for agreement between labellers.
- Tune the detector's thresholds by **grid search** on some sessions and score once on **held-out** ones (macro-F1) against an LLM labeller and an idle-time baseline; added progress logs that separate struggle from stuck.

❖ **AI Digital Twin Agent | RAG, Tool Calling, Gradio, Hugging Face Spaces** *July 2026*

- Built and deployed a personal AI agent with a **RAG pipeline** (chunking, embeddings, vector store) and **parallel and sequential LLM tool calling**, hosted on Hugging Face Spaces with a Gradio UI.
- Added **production guardrails**: conversation-history management, system and user prompt separation, token and API cost tracking, and a tool that alerts me when a visitor asks to connect.

EDUCATION & CERTIFICATIONS

❖ **MEng, Industrial Engineering | University of Toronto, Canada** *September 2025 – Present*

- **Relevant Courses:** Deep Learning, Applications of NLP, Data Science & Statistical Learning
- **Emphasis:** Data Analytics & Machine Learning, ELITE | GPA: 3.8/4

❖ **B. Tech, Industrial Engineering | College of Engineering Trivandrum, India** *August 2017 – June 2021*

❖ **Certifications:** Google Generative AI Leader (2026) | AWS AI Practitioner (2024) | Azure Data Scientist Associate (2024) | AWS Cloud Practitioner (2023)